Towards quantisation of Boussinesq solitons: Phase 2 report (bad equation first)

Project *quantum_boussinesq_solitons*

3 November 2026

Abstract

Following the PI's decision, we quantise the *bad* Boussinesq equation first:

$$v_{tt} = v_{xx} + v_{xxxx} + 3(v^2)_{xx}.$$

Its soliton sector is regular and purely elastic, it has regular breathers, and it is the continuum limit of the quantum Toda chain (Phase 1).

The main result comes from Route A. Two-soliton scattering is an exactly symplectic map for the canonical structure of the field theory, but *only* once ordering-dependent "topological" cross terms $d\Pi_i \wedge dM_j$ are included. These terms come from the kink in the potential ϕ. With them we obtain a closed-form semiclassical two-soliton phase, $S \simeq e^{i\delta/\hbar}$:

$$\delta(1,2) = \tfrac{2}{9}(\ell_1 - \ell_2)\log A_{12} + \tfrac{2}{3}M_1M_2(v_1 - v_2), \qquad \ell = \epsilon\cosh 3\beta, \quad k = \sinh\beta.$$

- δ is antisymmetric (unitarity).
- It obeys a crossing-type relation under $\beta \to \beta + i\pi$.
- Continued to complex conjugate rapidities, $i\cdot\frac{2}{9}(\ell_1 - \ell_2)$ reproduces *exactly* the adiabatic invariant of the Boussinesq breather computed independently by Bogdan and Charkina [1].

Breather quantisation thus becomes a string condition on the spectral parameters.

Routes B, C and D give structural results (non-ultralocality; a factorisation of A_{12} into $\mathfrak{sl}_3$ Toda and sine-Gordon blocks of $\beta_1 \pm \beta_2$; the quantum Toda chain as a regularisation) but no exact S-matrix. *No exact quantum S-matrix is obtained.* We end with a ranked list of next steps.

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1 Setting

Throughout, $v = 2(\log\tau)_{xx}$ and $k = \sinh\beta$. A soliton has velocity $v = \epsilon\cosh\beta$, with $\epsilon = +1$ for right-movers and $\epsilon = -1$ for left-movers. The potential is $v = \phi_x$ and the Lagrangian is

$$\mathcal{L} = \tfrac{1}{2}\phi_t^2 - \tfrac{1}{2}\phi_x^2 + \tfrac{1}{2}\phi_{xx}^2 + \phi_x^3,$$

with $\pi = \phi_t$ and canonical symplectic form $\Omega = \int \delta\pi \wedge \delta\phi$.

On one soliton the charges are:

- mass $M = \int v = 2k$;
- ϕ-shift charge $\Pi = \int \pi = -vM$, conserved because $\phi \to \phi + c$ is a symmetry;
- momentum $P = -\int \pi\phi_x = \frac{2}{3}\epsilon k^3\cosh\beta$;
- energy $E = \frac{2}{15}k^3(5 + 4k^2)$.

These satisfy $dE/dP = v$. **(a)** [1]/**(c)**

The two-soliton constants are

$$A_{\mathrm{ov}} = \frac{1 + 3\coth^2\frac{\beta_1+\beta_2}{2}}{1 + 3\coth^2\frac{\beta_1-\beta_2}{2}}, \qquad A_{\mathrm{ho}} = \frac{1 + 3\tanh^2\frac{\beta_1-\beta_2}{2}}{1 + 3\tanh^2\frac{\beta_1+\beta_2}{2}},$$

both in $(0, 1)$, and the position shifts are $\Delta x_1 = -k_1^{-1}\log A$, $\Delta x_2 = +k_2^{-1}\log A$ (soliton 1 starts on the left). **(c)** (Phase 1)

What a soliton S-matrix would mean here. The model has no boost symmetry and is gapless (phonons with $\omega \simeq |k|$). The natural object is therefore the non-relativistic two-body S-matrix $S(P_1, P_2)$ of soliton asymptotic states, in the sense of Faddeev–Korepin [6]. Classical scattering is purely transmissive and diagonal, so S is a phase; for two species (left and right movers) S is diagonal.

For diagonal scattering the Yang–Baxter equation is automatic. The non-trivial requirements are:

- unitarity;
- the correct semiclassical limit;
- analyticity and a crossing substitute;
- a bound-state (breather) spectrum consistent with poles.

2 Route A: semiclassical quantisation

2.1 The symplectic structure of soliton scattering

Proposition 2.1 (**(c)**). *Consider two well-separated solitons, with ϕ normalised by $\phi(-\infty) = c$. Up to exponentially small terms,*

$$\Omega = d\Pi_{\mathrm{tot}} \wedge dc + \sum_j dP_j \wedge dX_j + \big[X_i > X_j\big]\, d\Pi_i \wedge dM_j.$$

The argument is as follows. π_i is localised at X_i, and there ϕ_j equals 0 or M_j according to whether soliton i is to the left or right of soliton j. The self-term $dP \wedge dX$ was derived in Phase 1. In sine-Gordon the analogous cross term vanishes, because the kink height $2\pi/\beta$ is fixed. Here $M = 2k$ varies with the soliton parameter.

Proposition 2.2 (**(c)**, `scripts/symplectic_scattering.py`)**.** *The classical scattering map $X_j \mapsto X_j + \Delta x_j$ (with k_j and c fixed) preserves Ω, i.e.*

$$\sum_j dP_j \wedge d\Delta x_j = (\textit{cross term})_{\text{in}} - (\textit{cross term})_{\text{out}}.$$

This was verified to 40 digits for overtaking and head-on collisions at five parameter points each. It was also verified for the good *equation in its regular region (`scripts/consistency_checks.py`). Without the cross terms the identity fails by $O(1)$ amounts; for example, the left side equals* 19.79 *at $\beta = (0.9, 0.3)$.*

This is a non-trivial confirmation of the classical picture. It also shows that a semiclassical S-matrix built from the naive rule "$\Delta x_j = -\partial\delta/\partial P_j$" does not exist: the shifts are not curl-free in the canonical momenta.

2.2 The semiclassical phase

In each ordering region we choose Darboux coordinates by the rule

$$X_j' = X_j + \gamma_j \sum_{i \text{ left of } j} M_i, \qquad \gamma_j = \frac{d\Pi_j/dk_j}{dP_j/dk_j}.$$

The mirrored rule (normalising ϕ at $+\infty$) gives the same in–out differences. The shifts $\Delta x_j'$ are then curl-free. Their potential is the semiclassical phase δ, with $\Delta x_j' = -\partial\delta/\partial P_j$ and $S \simeq \exp(i\delta/\hbar)$, by stationary phase on two-particle wave packets [6].

Proposition 2.3 (**(c)**, `scripts/semiclassical_phase.py`, `scripts/consistency_checks.py`)**.**

1. $\dfrac{dP}{d\beta} = \frac{2}{3}\,k\,\sinh 3\beta$ *for the bad equation. For the good equation,* $\dfrac{dP}{d\alpha} = \frac{2}{3}\,k\sin 3\alpha$ *with $k = \sin\alpha$; this vanishes at the cusp $\alpha = \pi/3$ found in Phase 1.*

2. *For both overtaking and head-on collisions, and both equations,*

$$\boxed{\delta(1,2) = \tfrac{2}{9}\,(\ell_1 - \ell_2)\,\log A_{12} + \tfrac{2}{3}\,M_1 M_2\,(v_1 - v_2)} \tag{1}$$

   with $\ell = \epsilon\cosh 3\beta$ (bad) or $\ell = -\epsilon\cos 3\alpha$ (good). The remainder was first found by symbolic integration, then identified as $\frac{2}{3}M_1M_2(v_1 - v_2) = -\frac{2}{3}(\Pi_1 M_2 - \Pi_2 M_1)$ by exact matching.

3. Unitarity: $\delta(1,2) = -\delta(2,1)$.

4. Crossing-type relation: $\delta_{\text{ov}}(\beta_1, \beta_2 + i\pi) = -\delta_{\text{ho}}(\beta_1, \beta_2)$. *The shift $\beta \to \beta + i\pi$ maps a right-mover to a left-mover with $M \to -M$.*

Remark 2.4.

- ℓ is proportional to the spectral parameter of the third-order Lax operator. In the uniformisation of [4], $\lambda = \frac{1}{2}(k^3 + k^{-3})$, with k^3 playing the role of $e^{3\beta}$.
- The dynamical part of δ is (difference of spectral parameters) $\times \log A$.
- The second part is a "mutual statistics" phase: an Aharonov–Bohm-like phase $\propto \Pi_1 M_2 - \Pi_2 M_1$ between objects carrying ϕ-shift charge and ϕ-winding. **(d)** (interpretation)

Remark 2.5 (KdV limit, **(c)**). For small amplitudes the topological term is of the *same order* as the KdV-type term $2\beta_1 \log A$ in $\partial\delta/\partial\beta_1$. Numerically, their ratio tends to 0.564 for $\beta_1/\beta_2 = 1.7$. The heuristic of Bonnemain and Doyon [2] identifies the quantum phase derivative with $P'_{\text{GHD}}\Delta$, with $P_{\text{GHD}} = \eta^2/2$. That identification reproduces our dynamical term but *misses the topological term.* The classical generalised hydrodynamics, which uses only Δx, is unaffected. **(c)**

2.3 Bohr–Sommerfeld: breathers as spectral-parameter strings

Bogdan and Charkina quantise the Tajiri–Murakami breather of the bad equation by Bohr–Sommerfeld, $I_{\mathrm{br}} = N\hbar$, with an independently computed adiabatic invariant I_{br} **(a)** [1, (64)–(65)].

Proposition 2.6 (**(c)**). *Set $\beta_{1,2} = a \pm ib$, which is Bogdan–Charkina's parametrisation $\kappa \pm ik = \sinh(a \pm ib)$. Then:*

1. *A_{12} equals their a_{12} [1, (33)]. Regularity $A > 1$ holds if and only if $b > \pi/3$, which is their existence boundary.*
2. *$i \cdot \frac{2}{9}(\ell_1 - \ell_2) = I_{\mathrm{br}}$ exactly, with $I_{\mathrm{br}} = -\frac{4}{9}\sinh 3a \sin 3b$, in agreement with [1, (64)].*

Hence the coefficient of $\log A$ in (1) is the breather action. Bohr–Sommerfeld quantisation reads

$$\ell_1 - \ell_2 = -\tfrac{9i}{2}\, N\hbar, \qquad N \in \mathbb{N}, \tag{2}$$

which is a string condition in the spectral parameter ℓ.

This is the strongest cross-check in this report. It ties our two-soliton phase, built from Hirota shifts and the canonical symplectic form, to an independent quasiclassical calculation in the literature. Item 2 was checked at three parameter points to 10^{-16}.

Equation (2) is the form in which bound states appear in Bethe-ansatz solvable models. It suggests that ℓ is the variable in which an exact S-matrix has poles: $\ell_1 - \ell_2 \in -\frac{9i}{2}\hbar\mathbb{N}$, possibly with $\hbar$ renormalised. **(d)**

2.4 What Route A gives and where it stalls

- *Obtained.* The leading semiclassical phase (1), which is unitary, crossing-type covariant and consistent with breather quantisation. Also an asymptotic Bethe-ansatz quantisation of N solitons on a ring, $P_j\Lambda + \sum_{k\neq j}\delta(j,k) = 2\pi\hbar n_j$. **(d)**: the twisted boundary conditions forced by ϕ-winding on a ring have not been worked out.
- *Stalls at* the $O(\hbar^0)$ (one-loop) corrections, which need the soliton fluctuation spectrum and a renormalisation of $\hbar$. Without them the leading phase cannot be promoted to an exact S-matrix by consistency alone. Unlike in the relativistic Toda case [11, 12], δ is *elementary*: no dilogarithm appears, because $dP \propto k\sinh 3\beta\, d\beta$ and integration by parts closes. Johnson's "q-deformation of the dilogarithm" step therefore has no direct analogue.

3 Route B: quantum inverse scattering on the line

What would have to be true. The Lax matrix $U(x,\lambda)$ would need an ultralocal r-matrix Poisson algebra, so that it can be regularised on a lattice and quantised à la Faddeev–Korepin–Sklyanin. Then a Bethe ansatz over a pseudo-vacuum with soliton excitations would give S.

Proposition 3.1 (**(c)**, `scripts/routes_BC_checks.py`). *In canonical variables the Lax data of Phase 1 are $q = \frac{3}{2}\phi_x - \frac{1}{4}$ and $r = \frac{1}{2}q_x + \frac{\sqrt{3}}{4}\pi$ (good-equation real form; the bad one differs by i's). Their brackets are*

$$\{q(x), r(y)\} = \tfrac{3\sqrt{3}}{8}\,\delta'(x-y), \qquad \{q,q\} = \{r,r\} = 0.$$

So the companion-matrix Lax connection is non-ultralocal.

Where it stalls. There are two known ways out, and neither yet gives a theory on the line.

1. Maillet's r/s formalism with lattice regularisation [13]. No construction exists for Boussinesq.

2. The Miura (free-field) map. This is the route of Bazhanov–Hibberd–Khoroshkin [3] and Hikami [9]. It yields $U_q(\widehat{\mathfrak{sl}}_3)$ T- and Q-operators, but in the chiral conformal setting on a circle, with no particle sector.

Dimakis et al.'s Yang–Baxter maps [5] concern polarisations in the matrix case and are vacuous for the scalar equation. The "beyond Yang–Baxter" remark there is therefore not an obstruction here. For diagonal scattering, Yang–Baxter holds trivially. **(c)**

4 Route C: W_3, sl_3 affine Toda and the $U_q(\widehat{\mathfrak{sl}}_3)$ R-matrix

What would have to be true. Boussinesq solitons would have to be identifiable as a limit of $a_2^{(1)}$ (complex) affine Toda solitons, whose S-matrices are the $U_q(\widehat{\mathfrak{sl}}_3)$ intertwiners of Hollowood [11] and Gandenberger [7].

Proposition 4.1 (**(c)**, `scripts/routes_BC_checks.py`)**.** *Take Hollowood's $a_2^{(1)}$ interaction constant [10, (1.19)], $A^{(12)}(\theta) = \frac{2\cosh\theta+1}{2(1+\cosh\theta)}$, between the two fundamental species. With $S = \beta_1+\beta_2$ and $D = \beta_1-\beta_2$,*

$$A_{\mathrm{ov}}^{\mathrm{bad}}(\beta_1,\beta_2) = \frac{A^{(12)}(S)}{A^{(12)}(D)}\cdot\frac{\tanh^2(D/2)}{\tanh^2(S/2)}, \qquad A_{\mathrm{ho}}^{\mathrm{bad}}(\beta_1,\beta_2) = \frac{1}{A_{\mathrm{ov}}^{\mathrm{bad}}(\beta_1,\beta_2+i\pi)}.$$

Here $\tanh^2(\theta/2)$ is the sine-Gordon soliton–soliton constant. Equivalently,

$$A_{\mathrm{ov}} = \frac{G(S/2)}{G(D/2)}, \qquad G(y) = \frac{16\prod_{a=1,2}\sinh\frac{y+ia\pi/3}{2}\sinh\frac{y-ia\pi/3}{2}}{\sinh^2 y}.$$

So the Boussinesq interaction is built from exactly the $\mathfrak{sl}_3$ blocks, but in the form "$F(\beta_1-\beta_2)/F(\beta_1+\beta_2)$". This is the structure of scattering off a particle *and its mirror image*. It suggests that a Boussinesq S-matrix might take the form

$$S(\beta_1,\beta_2) = \mathcal{S}(\beta_1-\beta_2)\,\mathcal{S}(\beta_1+\beta_2+i\pi)^{\pm1}$$

with $\mathcal{S}$ built from $U_q(\widehat{\mathfrak{sl}}_3)$ and sine-Gordon factors. Such a form is reminiscent of boundary (reflection) amplitudes. That would connect to the "boundary affine Toda" structure in Bazhanov–Hibberd–Khoroshkin [3]. **(d)**

The obstruction is that the semiclassical phase (1) is not of this product form. Its weight $\frac{2}{9}(\ell_1-\ell_2)$ is not a sum of functions of S and D, and its topological part has no Toda analogue.

Further obstructions:

- Toda solitons are relativistic with fixed masses m_a. In the Toda constant the mass term $4m^2\sin^2(\pi a/3)$ appears where Boussinesq has K^4, so the species label is "replaced" by the wavenumber. There is no limit of fixed-species Toda solitons that yields Boussinesq solitons. **(c)**

- Takács and Watts show that $a_2^{(1)}$ soliton–breather scattering is flawed both classically and quantum mechanically **(b)** [19].

Route C stalls: the structural similarity is precise, but no dictionary is available. What is lost in the non-relativistic reduction is the difference form, and with it the standard crossing.

5 Route D: discrete models

(i) Quantum discrete Boussinesq (Hikami [9]). This is the Belov–Chaltikian lattice W_3 chain, periodic, with a Baxter Q-operator. A soliton/particle sector would require a thermodynamic limit with an identified ground state and excitations. That has not been done in the literature, and we did not attempt it. *Stalls immediately*; it is a research project in itself.

(ii) Quantum Toda chain. This is our recommended route. The required ingredients are:

1. The exact quantum Toda chain **(a)** [8, 16] and its asymptotic Bethe ansatz **(a)**/**(b)** [18, 14, 17]. Its particle excitations become solitons in the classical limit **(b)** [14, 15].

2. The dressed particle–particle S-matrix in the finite-density ground state. This is computable from linear integral equations. *Not done.*

3. Its classical limit should reproduce the Toda soliton phase, obtained by the symplectic construction of §2 applied to the lattice. The lattice has the same structure: displacement $x_n \leftrightarrow \phi$, $p_n \leftrightarrow \pi$, compression $\sum r_n \leftrightarrow M$, $\sum p_n \leftrightarrow \Pi$. So the same cross terms arise. **(c)** (structure only)

4. The small-amplitude scaling limit should give (1). Phase 1 verified $\log A^{\mathrm{Toda}} / \log A^{\mathrm{bad}} \to 1$. The weight and topological terms remain to be compared.

If steps 2–4 succeed, the Toda dressed S-matrix in the scaling limit *is* a candidate exact quantum Boussinesq soliton S-matrix, because it carries a definite $\hbar$-dependence. **(d)**

6 Consistency checks on the candidate

The only candidate is the semiclassical phase (1).

Condition	Result	Status
Unitarity	$\delta(1,2) = -\delta(2,1)$, exact	**(c)** passed
Crossing (substitute)	$\delta_{\mathrm{ov}}(\beta_1, \beta_2 + i\pi) = -\delta_{\mathrm{ho}}(\beta_1, \beta_2)$	**(c)** passed
Yang–Baxter / factorisation	diagonal scattering, so YBE is trivial; classical shifts pairwise additive (Hirota condition, $N \leq 4$)	**(c)** passed
Semiclassical limit vs. time delays	by construction, with the symplectic consistency of Prop. 2.2	**(c)** passed
Bound states / bootstrap	breather existence boundary and action reproduced (Prop. 2.6); no exact pole structure, so the bootstrap cannot be tested	**(c)** partial
Good equation	formula (1) holds in the regular region; behaviour at the cusp and resonances not analysed	**(c)** partial

7 Conclusions and ranked open problems

Conclusions.

1. No route yields an exact soliton S-matrix. The furthest-developed result is the semiclassical phase (1). **(c)**

2. The topological cross term is unavoidable. Any quantisation must treat the ϕ zero mode and the kink charge M (a continuous "winding") properly. This is absent from sine-Gordon and Toda, and is the most important structural difference we found. **(c)**

3. The spectral parameter $\ell \propto \lambda$ organises everything: the phase weight, the breather action and the string condition. **(c)**/**(d)**

Ranked next steps, with reasons.

1. **Quantum Toda dressing in the scaling limit** (Route D, steps 2–4). This is the only route with an exactly solvable quantum theory whose classical limit is our model. It would test (1), including the topological term, and provide $\hbar$-exact data.

2. **One-loop corrections for breathers and solitons** (Dashen–Hasslacher–Neveu-style): the fluctuation spectrum around a soliton of the bad equation. This would fix the $\hbar$ renormalisation in (2) and test the "string" interpretation. It is feasible because the linearised operators are reflectionless (integrability).

3. **An exact ansatz in the variables μ (roots of the cubic).** A_{12} is a KP cross-ratio of the roots (verified, `scripts/consistency_checks.py`), so a product of Γ-function factors in root differences is a natural candidate. Test it against (1), the string condition (2) and unitarity.

4. **The mirror-image structure** of Prop. 4.1 and the possible link to boundary affine Toda [3].

5. **The good equation**: semiclassics near the cusp $\alpha = \pi/3$ (where $dP/d\alpha = 0$) and near the resonance lines (where $\log A \to \pm\infty$). Do the attractive head-on channels give quantum bound states that are absent classically?

6. **Non-ultralocal quantum inverse scattering** (Maillet) for the Boussinesq Lax pair. This is hard, and we place it lowest.

What changed in our understanding. We expected the main difficulty to be the singular or resonant solutions. For the bad equation the real issue is instead the topological, ordering-dependent part of the symplectic structure. We also found that the semiclassical data are far more rigid than expected: the breather action match is a non-trivial check.

Least sure.

- The physical status of the Darboux convention. We showed that the two natural conventions agree, but have not classified all allowed ones.
- Whether $S \simeq e^{i\delta/\hbar}$ is complete at leading order. A Maslov-type constant phase could be missing.
- The ring quantisation with ϕ-winding.

A Scripts

All are in `phase2/scripts`; each ends with `ALL CHECKS PASSED`.

- `symplectic_scattering.py`: Prop. 2.2.
- `semiclassical_phase.py`: derivation of $\nabla\delta$ and of the closed form.
- `fit_R.py`: hyperbolic form of the remainder.
- `consistency_checks.py`: Props 2.3–2.6, the cross-ratio, the KdV-limit finding, and the good equation.
- `routes_BC_checks.py`: Prop. 4.1 and the non-ultralocal brackets.

Failures along the way.

- The direct symbolic simplification of the symplectic identity timed out, so we switched to 40-digit numerics.
- The good-equation phase first used the wrong sign of ℓ, and the test failed until it was corrected.
- A tolerance set below double precision caused spurious failures.
- The KdV-limit comparison with Bonnemain–Doyon *does not* agree. This is recorded as a finding, not a passed test.

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